

		ITER, SIKSHA 'O' ANUSANDHAN (Deemed to be University)		Assignment 2																										
Branch		ALL		Programme																										
Course Name		Supply Chain Management Strategy		Semester																										
Course Code		OPL4004		Academic Year																										
				B. Tech.																										
				8 th																										
				2025-26																										
Assignment: 2			Topic: Operations management, Sourcing																											
Learning Level (LL)		L1: Remembering	L3: Applying	L5: Evaluating																										
		L2: Understanding	L4: Analysing	L6: Creating																										
Q's	Questions				COs	LL																								
1	“Service design involves physical, psychological, and sensory elements.” Analyse how these three elements interact to influence overall customer satisfaction.				CO3	L3																								
2	Analyse the operations management (OM) decisions designed to create a transformation process in your university.				CO3	L3																								
3	ABC Ltd., a small manufacturing unit, provides the following data for the year ended 31 st March 2025. Sales unit = 25,000; sales value = ₹4,00,000; fixed cost = ₹ 1,20,000; variable cost = ₹ 8 per unit. By using this information, (a) estimate the break-even point quantity, and (b) analyse the effect of the following cases on this estimated break-even quantity. (i) variable cost increases by 10%, keeping all other factors constant. (ii) selling price decreases by 10%, keeping all other factors constant. (iii) fixed cost increases by ₹ 30000, keeping all other factors constant.				CO3	L4																								
4	Consider a small clinic where functional groupings are created, such as medicine, pharmacy, laboratory, and X-ray. The patients, the product in this case, move between departments based on their need. Do you think that process layout is the best facility layout in this case. Justify your answer.				CO3	L3																								
5	Given the following manufacturing precedence relationships, compute the cycle time per day and the minimum number of workstations assuming eight working hours a day and 400 products need to be produced per day. <table border="1"><thead><tr><th>TASK</th><th>DESCRIPTIOPN</th><th>TASK TIME (Seconds)</th><th>PRECEDING TASK</th></tr></thead><tbody><tr><td>A</td><td>Punching</td><td>30</td><td></td></tr><tr><td>B</td><td>Bending</td><td>20</td><td>A</td></tr><tr><td>C</td><td>Combining</td><td>30</td><td>B</td></tr><tr><td>D</td><td>Painting</td><td>40</td><td>C</td></tr><tr><td>E</td><td>Polishiong</td><td>60</td><td>D</td></tr></tbody></table>				TASK	DESCRIPTIOPN	TASK TIME (Seconds)	PRECEDING TASK	A	Punching	30		B	Bending	20	A	C	Combining	30	B	D	Painting	40	C	E	Polishiong	60	D	CO3	L3
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6	A procurement manager has two suppliers before him to purchase a key component: (A) Supplier Q, a global supplier, with a purchase price of \$9,000 and transportation cost of \$1,500 (B) Supplier R, a local supplier, with a purchase price of \$10,200 and transportation cost of \$300 The procurement manager sought your expert advice. Analyse the case and suggest your				CO4	L4																								

	advice to the procurement manager.		
7	A mid-sized manufacturing company has traditionally sourced its raw materials from international suppliers due to lower costs. However, recent disruptions in global supply chains and growing community pressure have encouraged the firm to consider shifting to local suppliers. Analyse the risks and benefits of continuing with global sourcing versus transitioning to local sourcing and the factors the company should consider before making a final decision?	CO4	L3
8	While Apple Inc. designs its products like the iPhone in-house, it outsources much of its manufacturing to companies in China and India. Analyse the risks and benefits associated with this outsourcing decision taken by Apple Inc.	CO4	L3
9	“The purchasing strategy should support the business strategy of the firm to meet the firm’s long-term requirements”. Analyse this statement with an example of your choice.	CO4	L3
10	Hermit’s have a restaurant with an annual cost of goods sold of \$520000. Calculate the average inventory assuming 52 weeks per year and weeks of supply equal to 5.	CO4	L3

Note:

1. Assignment-II carries a weightage of 6 marks.
2. The course outcomes 3 and 4 were covered.

Course Outcomes	CO3	Students will be able to analyse the process of integration, operations management, and marketing interfaces required to design efficient and responsive supply chain systems.
	CO4	Students will be able to analyse sourcing and logistics decisions through selection of supplier, transporter, warehousing, and distribution, using cost, service, and risk considerations.